

# Hanning Lu

lhnjames.github.io | Google Scholar

Efficient and reliable LLM systems, agent reinforcement learning, and inference optimization.

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Leeds, United Kingdom

## EDUCATION

### University of Leeds

*B.S. in Computer Science; expected first-class honours (1:1).*

Leeds, United Kingdom

*Sep 2024 – Jun 2027 (Expected)*

## RESEARCH EXPERIENCE

### Compiler Optimization and LLVM IR Decompilation

*Research Assistant | Supervised by Chunwei Xia, Zheng Wang*

University of Leeds, UK

*May 2025 – Sep 2026*

- Built a **compiler agent** that plans, executes, profiles, and revises optimizations using runtime and compiler feedback. [ACM ICS 2026 Workshop](#).
- Automated LLVM pass selection, parameter changes, and configuration tuning with correctness checks, achieving **2.25×** geometric mean speedup over -O3 on PolyBench (first-author preprint).
- Applied **iterative LLM self-correction** to LLVM IR decompilation, raising accuracy from ~50% to **over 90%** on obfuscated IR tests.
- Evaluated prompt framing in LLM cascades conditioned on ground truth, showing that more frequent revision does not imply more reliable revision (co-first-author preprint).

### Agent Self-Improvement and Reinforcement Learning

*Research Assistant | Supervised by Bin Chong*

Peking University, China

*Oct 2025 – Jul 2026*

- Proposed **LJ-GQA**, a **gradient-guided agent self-improvement method** that reweights advantages using output-gradient agreement across semantically similar decisions from independent rollouts.
- Achieved **86.5% success** on ALFWorld with Qwen2.5-1.5B; trajectory audits identified exploration-loop deadlocks as the remaining failure mode (in submission).
- Combined a **verifiable evidence graph** with Qwen3-0.6B LoRA-SFT and Active-GRPO for social bot detection: 98.9% / 98.8% accuracy on Fox8-23 / BotSim-24 and near-100% evidence coverage.

### Efficient LLM Inference and Model Optimization

*Research Assistant | Supervised by Tong Yang*

Peking University, China

*Jun 2024 – Jun 2025*

- Developed **adaptive token retention** across attention heads during generation, reducing KV-cache size by **70%** with ~1% performance loss on Llama-2-7B/LongBench.
- Studied quantization, structured/unstructured pruning, and sparse-weight inference for memory-efficient, low-latency deployment; co-authored a [quantization survey](#).

### TPS-Bench: Multi-Tool Agent Planning and Scheduling

*Research Assistant | Supervised by Zhijie Deng*

Shanghai Jiao Tong Univ.

*Jun 2025 – Aug 2025*

- Contributed to **TPS-Bench**, an MCP-based benchmark and agent framework for task decomposition, tool selection, and coordinated multi-step execution.
- Used **adaptive tool reselection**, execution scheduling, and intermediate-result aggregation to improve multi-tool planning reliability and fault recovery. [Code](#).

### Loop Transformers and Recurrent-Compute Reasoning

*Research Assistant | Supervised by Menglin Yang*

HKUST (Guangzhou)

*May 2026 – Sep 2026*

- Compared Frozen Loop, equal-magnitude noise, geometric steering, LoRA, and LoRA Loop on WinoGrande, bAbI, and ProofWriter; frozen-layer repetition did not consistently outperform noise.
- Used these controlled comparisons to identify **learned task-aligned recurrent updates** as the effective mechanism, rather than simply repeating frozen layers.

### Knowledge Graph Reasoning and Voice AI

*Research Assistant | Supervised by Jun Wei*

Tsinghua University, China

*Jun 2025 – Aug 2025*

- Combined **structured graph traversal** with cross-node evidence aggregation to support multi-hop knowledge-graph reasoning for complex question answering.
- Integrated LLM inference, incremental generation, streaming TTS, and audio delivery into a serving pipeline for **low-latency, real-time voice interaction**.

## INTERNSHIP EXPERIENCE

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### Xingjie Intelligence

Video Generation Infrastructure Engineer Intern

AI Startup

Jun 2026 – Sep 2026

- o Built **SGLang-based serving** for Wan diffusion models using tensor, pipeline, and sequence parallelism, KV caching, and block-wise autoregressive generation.
- o Profiled GPU bottlenecks with Nsight Systems, Nsight Compute, and PyTorch Profiler to guide optimization.
- o Evaluated FP8, weight-only/activation quantization, CUDA Graphs, kernel fusion, continuous batching, and streaming inference using FPS, chunk latency, TTFT, memory use, and scaling efficiency.

### Sheet0

Artificial Intelligence Engineer Intern

AI Startup

Jun 2025 – Sep 2025

- o Designed **multi-agent coordination** and configurable task-execution workflows to improve data-collection accuracy, runtime stability, and fault recovery.

## SELECTED PUBLICATIONS

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1. **H. Lu**, C. Xia. Agent-Agnostic End-to-End C/C++ Application Performance Optimization. Proceedings of the 40th ACM International Conference on Supercomputing Workshops, 2026. DOI: [10.1145/3774895.3812199](https://doi.org/10.1145/3774895.3812199).
2. W. Zhong\*, **H. Lu**\*, Z. Yao, S. Zhang, C. Xia. Revision Propensity Is Not Revision Reliability: Ground-Truth-Conditioned Evaluation of Prompt Framing in LLM Cascades. Preprint, 2026.
3. Y. Wang, T. Yang, X. Liang, G. Wang, **H. Lu**, Z. Xu, Y. Li, W. Li. Art and Science of Quantizing Large-Scale Models: A Comprehensive Overview. arXiv preprint, 2024. [arXiv:2409.11650](https://arxiv.org/abs/2409.11650).
4. H. Jia, Y. Liu, B. Chong, Y. Yang, Y. Chen, J. Liang, Q. Li, **H. Lu**, K. Xu, H. Zheng, C. Zhang, H. Peng, P. S. Yu. FinHarness: An Inline Lifecycle Safety Harness for Finance LLM Agents. arXiv preprint, 2026. [arXiv:2605.27333](https://arxiv.org/abs/2605.27333). (in submission).
5. **H. Lu**, Y. Yang, J. Su, Y. Liu, Z. Yao, Y. Li, T. Liang, Z. Zhang, R. Ran, K. Xu, B. Chong. SAHG: Sector-Anisotropic Hyperbolic Graph Model for Social Bot Detection. arXiv preprint, 2026. [arXiv:2605.30166](https://arxiv.org/abs/2605.30166). (in submission)
6. **H. Lu**, K. Hu, Y. Yang, Y. Liu, Z. Yao, Y. Li, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, B. Chong. VeriBot: Evidence-Routed Social Bot Detection with Auditable Decision Chains. (In submission)
7. K. Hu, Y. Yang, **H. Lu**, Y. Liu, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, H. Peng, P. S. Yu, B. Chong. BotRoute: Dual-Branch Selective Routing for Social Bot Detection. (In submission).
8. Z. Yao, Y. Tang, **H. Lu**, Y. Li, M. Yang. Hierarchy-Aware Sparse Autoencoders via Activation-Cone Routing. (In submission).
9. Z. Yao, **H. Lu**. HCGM: Hyperbolic Code Graph Modeling for Repository-Level Software Intelligence. (In submission).

\* Co-first authors.

## COMPETITIONS & HONORS

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**Gold Medal**, CCPC Zhengzhou Invitational – Team Captain

2024

**Silver Medal**, 48th ICPC Xi'an Invitational – Team Captain